The Balancer and Audio Settings

Every audio setting Node Talker keeps, its range and default, which settings only take effect when the level settings are turned on, how the automatic balancer changes them, and the limits that stop a test when the Pi is under strain.

SETTINGS · LEVEL SWITCH · BALANCER · LOAD GUARD

SUMMARY · READ-ONLY REVIEW

Every audio setting

SETTING	DEFAULT	ALLOWED	WHAT IT DOES
`tune_enabled`	off	on or off	The level switch (Part 2). Gain, listen buffer and transmit buffer take effect only when it is on
`rx_gain_db`	0	-12 to +12 dB	Fixed gain on the receive audio, added to the AGC gain
`tx_gain_db`	0	-12 to +12 dB	Gain on the transmit audio
`jitter_ms`	250	100 to 1500 ms	The listen buffer on the page
`tx_buffer_ms`	120	40 to 400 ms	How much audio is held before transmit starts
`rx_hpf`	on	on or off	The 300 Hz high-pass filter (RX series, Part 3)
`rx_agc`	on	on or off	Automatic gain on the receive audio
`rx_gate`	1	0, 1 or 2	The receive gate (off, gentle, strict)
`rx_notch_hz`	0 (off)	0, or 60 to 300 Hz	A notch filter for one tone
`kerchunk_delay_ms`	0 (off)	0, or 100 to 1000 ms	Holds back audio so short kerchunks are dropped
`busy_lockout`	on	on or off	Refuses a key while the channel is busy
`tot_sec`	120	30 to 600 s	The longest transmission before timeout
`ptt_mode`	hold	hold or tap	How the page's push-to-talk button works
`ptt_guard_sec`	60	30, 60, 90, 120 or 180 s	When the page warns that a transmission is long (never beyond the time limit)
`ptt_rx_gap`	off	on or off	The page waits for a gap in the received audio before it keys
`phone_tones`	on	on or off	Opens a phone session on each key-up (TX path, Part 7)
`phone_open` / `phone_close`	*99 / #	1 to 8 of 0-9, * and #	The phone tone codes
`phone_lead_ms`	500	0 to 3000 ms	How long the phone session stays open after the close code
`max_listeners`	3	1 to 10	Listener slots (RX series, Part 4)
`allow_wan`	off	on or off	Opens the gated routes to the internet

Values outside the allowed range are refused with a message that names the rule.

Why some settings have no effect

Gain, the listen buffer and the transmit buffer are read through a single rule. When `tune_enabled` is off, the server uses the defaults for those three settings, whatever is saved. The saved values take effect only when the level switch is on.

SETTING	LEVEL SWITCH OFF	LEVEL SWITCH ON
`rx_gain_db`	0 dB	Saved value
`tx_gain_db`	0 dB	Saved value
`tx_buffer_ms`	120 ms	Saved value
`jitter_ms`	Page uses 250 ms	Saved value

The filter, AGC, gate, notch, kerchunk and busy settings are read directly from the saved values, so they take effect whether or not the level switch is on.

When a balancer change turns the switch on, the other level settings are pinned to the values that were in effect while the switch was off. So the change is the only thing that is heard, not a jump to several saved values at once.

During a tone test, the server may apply a temporary gain. The temporary gain is cleared when the test ends, and the saved values are not changed.

How a balance run changes settings

The balancer runs in the page. It makes one change at a time, measures again, and keeps the change only if the score does not get worse. The server does not compute the changes; it saves and restores the settings.

3.1 Starting and undoing

1. **Snapshot.** The server saves the current balance settings (the level switch, RX and TX gain, RX filter and gate settings, the listen buffer and the transmit buffer) to a file, `nodetalker_balance.json`. If that file already exists, a new run is refused until the earlier one is restored or kept.
2. **Restore.** Writes the saved settings back and removes the file.
3. **Keep.** Removes the file and leaves the changed settings in place.

A test that is turned off under Tests (`tests_off`) cannot start a balance run.

3.2 The tone rounds (RX gain)

The node is asked for a tone test, and the tones are measured at each frequency. The score is based on the 1000 Hz tone (or the loudest tone if 1000 Hz is missing), which should come back at -16 dBFS:

- The difference from the target is measured. If it is within 1.5 dB, nothing changes for that tone.
- Otherwise the RX gain moves by the difference, limited to 6 dB a round, and kept between -12 and +12 dB.
- Dropouts count against the score at half weight. If tones drop out, the listen buffer moves up through 150, 250, 500 and 1000 ms, instead of the gain.
- The run makes up to three rounds. If a round scores worse than the one before by more than 0.5, that change is undone and the run stops.

3.3 The live rounds

During a live check, the balancer measures the operator's audio on both sides:

- **Receive score:** three times the clipping percentage, twice the drops, plus the resets and tails, plus how far the speech level is from -19 dBFS.
- **Transmit score:** half the distance of the speech level from -18 dBFS, plus three times the clipping percentage.

Before each change, the balancer waits for a quiet channel: five seconds with no audio and no transmission, up to a limit of two minutes. If the channel stays busy, the run stops. The live rounds also change only one setting at a time, and undo a change that made the score worse by more than 0.5.

The automatic mode is limited to these settings:

- **Receive:** `rx_gain_db`, `rx_hpf`, `rx_agc`, `rx_gate`, `jitter_ms`, and the app's own buffer where an app is used.
- **Transmit:** `tx_gain_db` and `tx_buffer_ms`.

3.4 Limits on the choice

The automatic mode is limited to the keys above. The live run chooses from the page's suggestion list, which was not traced in full.

Stopping a test when the Pi is strained

While a tone test runs, a watch checks the Pi each second. The test stops, with a message, when any of these holds:

CHECK	LIMIT
CPU use	Above 90 % for 10 seconds in a row
Free memory	Below the larger of 40 MB or 10 % of the total
Temperature	80 degrees C or more
Throttling	The Pi reports it is throttling

The watch also checks the link to the node at intervals. A test stops with a message if the node is no longer linked or ready.

What the balancer and settings show

5.1 Change one thing at a time

The balancer's rule is the important one for operators: one setting, one measurement, keep or undo. A tester who changes several settings by hand gets no such check, and the book should say so.

5.2 Read the level switch first

An operator who edits the gain and sees no change should check the level switch. The gain only takes effect when it is on.

5.3 A run can be left open

A balance run that is not restored or kept leaves its file behind, and blocks the next run. The page offers a restore. The book should tell operators to restore or keep a run before they start another.

5.4 Not checked

- The page's suggestion engine (the function that picks the settings for the live rounds), read only in part.
- The app's own buffer, which is changed by the app and not by the server.
- The exact tone measurement on a running node.